DSC 102 · DISCUSSION

MLE Challenges and PA3: Ventilator Pressure Prediction

How the two AI Coding Gym challenges work, and how to get started on PA3

PA3 due Nov 13, 11:59 PM Pacific

TODAY

Agenda

1 The two MLE challenge assignments

2 PA3: the problem, data, scoring and rules

3 Getting set up: DataHub GPUs, the gym tool, using AI, the report

THE BIG PICTURE

Two MLE challenges on AI Coding Gym

Each is a real machine learning competition where frontier AI agents fall short. Your goal: **outperform the AI** with your own judgment, careful experiments and AI as a tool.

Challenge	Problem	Dates
1 · PA3	Ventilator Pressure Prediction	Oct 26 – Nov 13
2 · [PA4]	[TBA]	[TBA]

Both challenges use the same workflow, deliverables and grading: the next two slides.

BOTH CHALLENGES

The learning loop

1

Inspect

Study the AI agents' runs on the challenge page (recommended)

2

Build

Solve it with AI as a tool; override it where it is weak

3

Submit

Your score appears on the live leaderboard

4

Share

2–3 students present work in progress in class

5

Reflect

Write the report: what the AI missed and what you learned

Repeat steps 2 and 3 as often as your time allows.

BOTH CHALLENGES

What you submit and how it is graded

Deliverables

1. Predictions on AI Coding Gym, plus your gym username on Canvas
2. Your full AI interaction history (Canvas)
3. A 3–5 paragraph report PDF (Canvas)

Grading

Component	Weight
Leaderboard score	30%
Reflection write-up	40%
AI diagnosis (your own AI)	30%

Understanding matters more than the raw score: missing the AI baseline with a perceptive reflection can still earn a strong grade.

CHALLENGE 1 · PA3

Ventilator Pressure Prediction

Due Friday, Nov 13, 11:59 PM Pacific · no late days

aicodinggym.com/challenges/mle/ventilator-pressure-prediction

PA3 · THE PROBLEM

Predict the airway pressure during each breath

A ventilator pumps air into a sedated patient’s lungs through a tube. Here, a modified open-source ventilator drives an artificial bellows test lung, and each breath lasts about 3 seconds.

1 · Breathe in: inspiratory phase

The inspiratory valve opens (u_in, 0 to 100%) to let air into the lung

Scored

2 · Breathe out: expiratory phase

The expiratory valve opens (u_out: 1 = open, 0 = closed) to let air out

Not scored

Inputs per time step

Control inputs u_in, u_out; lung attributes R, C; time_step

Your model

Learns from training breaths, across lungs with different attributes

Output

Airway pressure (cmH2O) at each time step

Three files, one row per time step

Files

File	Has pressure?
`train.csv`	Yes
`test.csv`	No
`sample_submission.csv`	Format only

Columns

Column	Meaning
id	Unique time step identifier across a file
breath_id	Unique identifier of a breath
R	How restricted the airway is (cmH2O/L/s)
C	How compliant the lung is (mL/cmH2O)
time_step	The actual time stamp
u_in	Inspiratory valve control input, 0 to 100
u_out	Expiratory valve control input, 0 or 1
pressure	Airway pressure (cmH2O): the target

Mean absolute error, inspiratory phase only

Score = MAE between predicted and actual pressures **during the inspiratory phase** of each breath.

The expiratory phase is not scored.

Lower is better.

Submission file

```
id,pressure
1,20
2,23
3,24
…
```

A header, and a pressure for **every id** in the test set

PA3 · LEADERBOARD

Where the AI agents landed

Reference point	MAE
Gold medal line	0.1119
Silver medal line	0.1338
Bronze medal line (top 10%)	0.1364
AI agent: MLEvolve	0.14237
AI agent: Disarray	0.15171
Leaderboard median	0.1638
AI agent: AIBuildAI	1.0759

The best agent beats the median but misses a medal.

Your best submission is ranked on the platform’s leaderboard.

PA3 · RULES

Use any AI, but not existing solutions

Allowed

- Any AI tools you like
- Any compute you have
- Ideas from your own data analysis, even if top teams found them too

Not allowed

- The original Kaggle data, or datasets and models built from it
- Existing solutions: notebooks, discussions, write-ups, repos
- Asking an AI what the winning teams did

Disclose

- Any competition-specific trick an AI suggests unprompted
- Any idea from an outside source, credited in your report

Use only the data from aicodinggym mle download: the hidden test set came from the original data.

SETUP · STEP 1

Start a GPU container on DataHub

1. Off campus? Connect to the UCSD VPN
2. Log in to the launch server (it only launches containers; don’t run code there)
3. Launch the course GPU environment; if GPUs are busy, wait in the queue, don’t press Ctrl-C
4. Open the printed Jupyter link with the full host name, dsmlp-login.ucsd.edu
5. In JupyterLab: File → New → Terminal, then check the GPU

```
# 2. log in
ssh <username>@dsmlp-login.ucsd.edu
# 3. launch (1 GPU, 8 CPU, 16 GB)
launch-scipy-ml.sh -W DSC102_FA26_001 -g 1 -c 8 -m 16
# 5. in the Jupyter terminal
nvidia-smi
```

SETUP · STEP 2

Download the data and submit with the gym tool

```
# install once

pip install aicodinggym-cli
echo 'export PATH=$HOME/.local/bin:$PATH' >> ~/.bashrc
export PATH=$HOME/.local/bin:$PATH
aicodinggym configure --user-id <your-user-id>

# get the data

aicodinggym mle download ventilator-pressure-prediction
cd ~/ventilator-pressure-prediction/data
unzip ventilator-pressure-prediction.zip

# submit (as often as you like)

aicodinggym mle submit ventilator-pressure-prediction \
-F submission.csv
```

Your user ID is on your AI Coding Gym profile page.

Configure once: your home folder is kept between containers.

The score prints right away and shows on the challenge page. Only your best counts.

SETUP · USING AI

Three ways to work with AI on DataHub

A

Chat in your browser

ChatGPT, Claude, Gemini and others. Paste code into your Jupyter notebooks or scripts.

Easiest to start.

B

Coding agent in the container

Claude Code, Codex CLI, Gemini CLI and others, run in the Jupyter terminal. It reads, edits and runs code on your GPU.

Most hands-off.

C

Your editor, connected

VS Code or Cursor on your laptop, connected to the container by Remote-SSH. AI extensions work on your DataHub files.

Familiar editor.

The container has internet access. You can also develop on your laptop with any AI tool and copy your code to DataHub only for training.

USING AI · OPTION B

Run a coding agent in the container

1. Start your GPU container and open a terminal in JupyterLab
2. Install the agent’s command-line tool, following its own docs
3. Log in to the agent, then start it in your project folder
4. Ask it to run long training in the background, with logs and checkpoints

```
# if a global npm install is not allowed,
# install into your home folder:
npm install -g --prefix ~/.local <package>
# start it where your project lives
cd ~/ventilator-pressure-prediction
```

The gym tool puts instruction files for AI agents in your home folder (AGENTS.md, CLAUDE.md, GEMINI.md, .cursorrules). Read them; agents started there read them too.

USING AI · OPTION C

Connect VS Code or Cursor to your container

1. On your laptop, create an SSH key with ssh-keygen, and add the public key to ~/.ssh/authorized_keys on dsmlp-login
2. Install the Remote-SSH extension and add this host to your laptop’s ~/.ssh/config
3. On the VPN if off campus, connect to dsc102-gpu: a GPU container starts, and your AI extensions work on its files

```
Host dsc102-gpu
User <username>
ProxyCommand ssh -i <path-to-private-key> <username>@dsmlp-login.ucsd.edu
/opt/launch-sh/bin/launch-scipy-ml.sh -W DSC102_FA26_001 -g 1 -H -N vscode-dsmlp
# the ProxyCommand is ONE line in the file; it is split here to fit
```

USING AI · WHICHEVER OPTION

Keep your AI history from day one

If you use…	Save
Chat in a browser	Every conversation: export it or save a share link as you go
A coding agent	Its session transcripts and logs, copied into your home folder
An editor’s AI	Its chat panels and the edits you accepted or rejected

Log your overrides

Each time the AI was wrong and you caught it: what, how, why. That is your AI diagnosis.

Tell your AI the rules

No existing solutions to this competition, and no original Kaggle data.

PLEASE REMEMBER

GPUs are limited. Start early.

The course GPU option shares a pool of GTX 1080 Ti GPUs (11 GB) with other DataHub users. You may wait in a queue, and waits grow near the deadline. Unless you have your own compute, do not leave training to the last days.

6 h limit

Containers stop after 6 hours (12 max with K8S_TIMEOUT_SECONDS)

Don’t lose work

Use -b for long runs and save checkpoints often

Free the GPU

kubectl delete pod <name> when you are done

PA3 · THE REPORT

Three to five paragraphs, four parts

AI inspection summary

What your own AI assistant did, its key decisions, and the errors and weaknesses you found

Your strategy

What you changed and why, the prompts you used, the judgment the AI lacked

Results comparison

Your score against the AI agents, with a leaderboard screenshot

Reflection

Where the AI fell short, what you did that it did not, what this taught you

Tip: save your AI chats and logs from day one, and note each time you caught your AI making a mistake.